

C889

Molecular and Cellular Biology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a lab funded by a regional grant, a researcher links 12 glucose molecules into one unbranched chain. The lab buys glassware from a supplier in another state. How many water molecules are released?

- A. None; carbohydrate bonds form without any water change.
- B. 11
- C. 11 water molecules are consumed, not released.
- D. 12

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An enzyme from a microbe found in a mountain lake is heated to 80°C for 10 minutes and stops working. The sample had been stored in a new freezer. Which level of the enzyme's structure remains intact?

- A. Tertiary structure
- B. Quaternary structure
- C. Primary structure
- D. All levels, since only activity was lost

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A mutation changes one amino acid in a protein. Which level of structure is changed directly, and why might the protein's function change?

- A. Primary structure; a new side chain can alter folding and the active site.
- B. Only quaternary structure; the chain itself is unchanged.
- C. Secondary structure changes first, which then rewrites the sequence.
- D. None; one amino acid is too small to matter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A pancreatic cell in a sample shipped overnight from a partner university makes insulin for export. The microscope used has a new camera. What is the correct path for insulin?

- A. Free ribosome → cytosol → plasma membrane
- B. Rough ER → Golgi → secretory vesicle → plasma membrane
- C. Golgi → rough ER → vesicle → plasma membrane
- D. Smooth ER → lysosome → plasma membrane

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A red blood cell whose cytoplasm is about 0.9% salt is placed in a 3% salt solution that a student technician prepared on a new balance. What happens to the cell?

- A. Water leaves the cell and it shrinks.
- B. Salt moves in until concentrations match, with no volume change.
- C. Nothing happens, because membranes never let water through.
- D. Water enters and the cell swells and bursts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A plant cell's central vacuole swells and the cell becomes firm (turgid). What was the surrounding solution relative to the cell?

- A. Isotonic
- B. Any solution, because the cell wall always pulls water in.
- C. Hypertonic
- D. Hypotonic

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A nerve cell keeps K⁺ at about 140 mM inside and 5 mM outside, a gradient first measured in studies of squid axons. The cell moves more K⁺ inward through a membrane protein. What kind of transport is this?

- A. Active transport, using energy such as ATP (the Na⁺/K⁺ pump).
- B. Simple diffusion across the lipid bilayer.
- C. Facilitated diffusion, because a protein is involved.
- D. Passive transport, since K⁺ naturally moves toward higher concentration.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A drug candidate from a startup with 40 employees binds the active site of an enzyme. Adding much more substrate restores the reaction rate to its original maximum. What type of inhibitor is it?

- A. Noncompetitive inhibitor
- B. Allosteric activator
- C. Competitive inhibitor
- D. Irreversible inhibitor

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

An inhibitor binds an enzyme at a site away from the active site. What happens to V_{max} when substrate concentration is raised greatly?

- A. V_{max} drops to zero.
- B. V_{max} stays lower than normal.
- C. V_{max} returns to normal.
- D. V_{max} rises above normal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A human cell (46 chromosomes) grown in a lab that recently upgraded its incubators finishes S phase. In G2, how many chromosomes and chromatids does it have?

- A. 23 chromosomes, 46 chromatids
- B. 46 chromosomes, 46 chromatids
- C. 46 chromosomes, 92 chromatids
- D. 92 chromosomes, 92 chromatids

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A human cell in G2 has 92 chromatids. After mitosis and cytokinesis, how many chromosomes does each daughter cell have?

- A. 46, each with two sister chromatids
- B. 92
- C. 23
- D. 46, each a single chromatid

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a mitochondrion studied by a lab that publishes in open-access journals, a chemical makes the inner membrane leaky to H^+ . What happens?

- A.** ATP synthesis rises because H^+ moves faster.
- B.** Glycolysis stops completely.
- C.** O_2 use stops, because the chain makes ATP directly.
- D.** ATP synthesis falls while electron transport and O_2 use continue, releasing energy as heat.

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

In a lab ~~funded by a regional grant~~, a researcher links 12 glucose molecules into one unbranched chain. ~~The lab buys glassware from a supplier in another state.~~ How many water molecules are released?

Target word: water released **Decides:** 12 monomers in a chain make 11 bonds.

A	Dropped element	Drops the dehydration step that forms every bond.
B	Correct	One water per glycosidic bond; $n - 1 = 11$.
C	Inverted operation	That describes hydrolysis, which breaks bonds.
D	Wrong stem number	Counts one water per monomer instead of per bond.

2. Answer C Concept 2

An enzyme from a microbe ~~found in a mountain lake~~ is heated to 80°C for 10 minutes and stops working. ~~The sample had been stored in a new freezer.~~ Which level of the enzyme's structure remains intact?

Target word: what survives denaturation **Decides:** Heat breaks weak interactions, not peptide bonds.

A	Confusion set	The 3-D fold is exactly what heat disrupts.
B	Adjacent category	Subunit associations rely on weak bonds and are also disrupted.
C	Correct	The amino acid sequence stays linked by covalent peptide bonds.
D	Absolute language	Activity is lost because shape changed.

3. Answer A Concept 2 • transfer

A mutation changes one amino acid in a protein. Which level of structure is changed directly, and why might the protein's function change?

Target word: cause runs from sequence to shape **Decides:** Sequence determines folding.

A	Correct	Primary structure dictates higher levels.
B	Adjacent category	The chain sequence is exactly what changed.
C	Inverted operation	Folding does not change the sequence; the arrow runs the other way.
D	Absolute language	Single substitutions can cause disease, such as in sickle-cell hemoglobin.

4. Answer B Concept 3

A pancreatic cell in a sample ~~shipped overnight from a partner university~~ makes insulin for export. ~~The microscope used has a new camera.~~ What is the correct path for insulin?

Target word: secretory route **Decides:** Insulin is secreted.

A	Adjacent category	Free ribosomes make proteins that stay in the cytosol.
B	Correct	The standard route for secreted proteins.
C	Inverted operation	Reverses the ER and Golgi steps.
D	Confusion set	Smooth ER makes lipids; lysosomes digest.

5. Answer A Concept 4

A red blood cell whose cytoplasm is about 0.9% salt is placed in a 3% salt solution ~~that a student technician prepared on a new balance~~. What happens to the cell?

Target word: direction of water movement **Decides:** The outside is hypertonic.

A	Correct	Water moves toward higher solute concentration outside.
B	Confusion set	Osmosis is water movement; the membrane limits salt.
C	Absolute language	Water crosses membranes, largely through aquaporins.
D	Inverted operation	That happens in a hypotonic solution.

6. Answer D Concept 4 • transfer

A plant cell's central vacuole swells and the cell becomes firm (turgid). What was the surrounding solution relative to the cell?

Target word: infer tonicity from the result **Decides:** Water moved in.

A	Adjacent category	No net water movement; the cell would be limp.
B	Absolute language	The wall resists swelling; it does not draw water.
C	Inverted operation	That would pull water out and cause plasmolysis.
D	Correct	Lower outside solute draws water into the cell.

7. Answer A Concept 5

A nerve cell keeps K^+ at about 140 mM inside and 5 mM outside, ~~a gradient first measured in studies of squid axons~~. The cell moves more K^+ inward through a membrane protein. What kind of transport is this?

Target word: transport type **Decides:** K^+ moves from low (5 mM) to high (140 mM).

A	Correct	Moving against a gradient requires energy.
B	Adjacent category	Ions cannot cross the bilayer directly, and this is uphill.
C	Confusion set	Facilitated diffusion only moves down a gradient.
D	Inverted operation	Passive movement runs from high to low.

8. Answer C Concept 6

A drug candidate ~~from a startup with 40 employees~~ binds the active site of an enzyme. Adding much more substrate restores the reaction rate to its original maximum. What type of inhibitor is it?

Target word: inhibitor type **Decides:** More substrate restores V_{max} .

A	Confusion set	Noncompetitive inhibition lowers V_{max} and is not overcome by substrate.
B	Inverted operation	An activator raises activity; the drug inhibits.
C	Correct	Substrate outcompetes it for the active site; V_{max} is unchanged.
D	Adjacent category	Permanent binding could not be reversed by substrate.

9. Answer B Concept 6 • transfer

An inhibitor binds an enzyme at a site away from the active site. What happens to V_{max} when substrate concentration is raised greatly?

Target word: V_{max} under noncompetitive inhibition **Decides:** Substrate cannot displace an inhibitor at another site.

A	Absolute language	Uninhibited enzyme molecules still work.
B	Correct	Inhibited enzyme molecules stay inactive regardless of substrate.
C	Confusion set	That is the competitive pattern.
D	Inverted operation	An inhibitor cannot raise maximum rate.

10. Answer C Concept 7

A human cell (46 chromosomes) ~~grown in a lab that recently upgraded its incubators~~ finishes S phase. In G₂, how many chromosomes and chromatids does it have?

Target word: counts after replication **Decides:** Replication doubles chromatids, not centromeres.

A	Adjacent category	Halving belongs to meiosis.
B	Dropped element	Drops the DNA replication of S phase.
C	Correct	Each chromosome now has two sister chromatids.
D	Confusion set	Counts chromatids as chromosomes.

11. Answer D Concept 7 • transfer

A human cell in G₂ has 92 chromatids. After mitosis and cytokinesis, how many chromosomes does each daughter cell have?

Target word: daughter chromosome number **Decides:** Sister chromatids separate; each daughter gets 46.

A	Confusion set	That is the G ₂ state before division.
B	Dropped element	Drops the separation into two cells.
C	Adjacent category	Meiosis, not mitosis, halves the number.
D	Correct	Mitosis keeps the parent's chromosome number.

12. Answer D Concept 8

In a mitochondrion ~~studied by a lab that publishes in open-access journals~~, a chemical makes the inner membrane leaky to H^+ . What happens?

Target word: effect of collapsing the H^+ gradient **Decides:** ATP synthase needs the H^+ gradient; electron transport does not need ATP synthase.

A	Inverted operation	H^+ bypasses ATP synthase, so less ATP is made.
B	Absolute language	Glycolysis is in the cytosol and does not need the gradient.
C	Confusion set	The chain pumps H^+ ; ATP synthase makes ATP.
D	Correct	The gradient leaks away before driving ATP synthase.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Dehydration synthesis and hydrolysis	Monomers join by dehydration synthesis (one water released per bond); hydrolysis adds water to break bonds.	Thinking building a polymer uses water, or counting one water per monomer.	A number of monomers linked into a chain.
2	Protein structure and denaturation	Primary structure (amino acid sequence) determines folding; heat or pH disrupts secondary, tertiary, and quaternary structure but leaves peptide bonds intact.	Believing denaturation breaks the amino acid chain.	An enzyme loses activity after heat or a pH change.
3	The secretory (endomembrane) pathway	Secreted proteins are made on rough ER ribosomes, modified in the Golgi, and shipped in vesicles to the plasma membrane.	Routing secreted proteins through free ribosomes, smooth ER, or lysosomes.	A cell that exports a protein such as a hormone or enzyme.
4	Osmosis and tonicity	Water moves across a membrane toward the side with more solute; cells shrink in hypertonic and swell in hypotonic solutions.	Thinking solute moves instead of water, or reversing the direction.	A cell placed in a solution with a stated salt or sugar percentage.
5	Active vs. passive transport	Moving a substance against its concentration gradient requires energy (active transport); facilitated diffusion uses a protein but no energy.	Calling any protein-assisted transport active.	Concentrations inside and outside the cell with a direction of movement.
6	Enzyme inhibition	Competitive inhibitors bind the active site and are overcome by more substrate (same V_{max}); noncompetitive inhibitors bind elsewhere and lower V_{max} .	Swapping the two types.	What happens when substrate concentration is raised.
7	Chromosomes vs. chromatids through the cell cycle	S phase doubles DNA and chromatids but not the chromosome count; mitosis gives two cells with the parent's chromosome number.	Doubling the chromosome number after S phase.	"How many chromosomes and chromatids in G2?"
8	Chemiosmosis and ATP synthase	The electron transport chain pumps H^+ into the intermembrane space; H^+ flowing back through ATP synthase makes ATP, with O_2 as the final electron acceptor.	Thinking the electron transport chain makes ATP directly.	A membrane made leaky to H^+ or a blocked ATP synthase.

Course scope (public description): Molecular and Cellular Biology provides graduate students seeking initial licensure or endorsement and/or to earn their MA degree in secondary science education with an introduction to the area of molecular and cellular biology. This course examines the cell as an organism emphasizing molecular basis of cell structure and functions of biological macromolecules, subcellular organelles, intracellular transport, cell division, and biological reactions. A prerequisite for this course is Introduction to Biology.